TIME SPEED AND DISTANCE

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- Speed, Time & Distance – Introduction & Concept
- Speed Distance Time is one of the most popular and important topics in the Mathematics or Quants section of any competitive exam. The concept of Speed, Time and Distance is used extensively for questions relating to different topics such as motion in a straight line, circular motion, boats and streams, races, clocks, etc. Aspirants should try to understand the inter-relationship between the factors speed, distance and time.
- Relationship Between Speed, Time & Distance
- **Speed = Distance/Time** – This tells us how slow or fast an object moves. It describes the distance travelled divided by the time taken to cover the distance.
- Speed is directly Proportional to Distance and Inversely proportional to Time. Hence,
- **Distance = Speed X Time**, and
- **Time = Distance / Speed**, as the speed increases the time taken will decrease and vice versa.

- **Average Speed = (Total distance traveled)/(Total time taken)**
- **Case 1 –** When the distance is constant: **Average speed = 2xy/x+y**; Where, x and y are the two speeds at which the same distance has been covered.
- **Case 2** – When the time taken is constant: **Average speed = (x + y)/2**; Where, x and y are the two speeds at which we traveled for the same time.
- Speed, Time & Distance Conversions :
- To convert from km / hour to m / sec, we multiply by 5 / 18. So, 1 km / hour = 5 / 18 m / sec
- To convert from m / sec to km / hour, we multiply by 18 / 5. So, 1 m / sec = 18 / 5 km / hour = 3.6 km / hour

TIME SPEED AND DISTANCE

Q 1. A train runs at the rate of 54 km an hour. What is its speed in metres per second?

(a) 15 m/sec (b) 25/2 m/sec (c) 8 m/sec (d) None of these

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TIME SPEED AND DISTANCE

Q 2. **A man is walking at a speed of 6 km per hour. After every km he takes rest for 6 minutes. How much time will he take to cover a distance of 18 km.**

(a) 5 hrs (b) 5 hrs 42 min (c) 4 hrs 42 min (d) None of these

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TIME SPEED AND DISTANCE

Q 3. A boy goes to school with 4/5th of original speed and he reached school 5 min. late. Find actual time?

(a) 8 min. (b) 16 min. (c) 20 min. (d) None of these

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TIME SPEED AND DISTANCE

Q 4. Speed of a father and a son is 12 km/h and 18 km/hr. They want to travel A to B. Son start after 39 min of his father start and reached 21 min. before father at B. Find distance b/w A to B?

(a) 18 km (b) 36 km (c) 42 km (d) None of these

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TIME SPEED AND DISTANCE

Q 5. A person goes to his in-laws house at 40 km/hr he reach 2 hrs early and next time he goes with 30 km/h, he reach 1 hour earlier. Find distance, actual time and speed?

(a) 120 km,5h,24km/h (b) 150km,6h,30km/h (c) 180km,8h,35km/h

(d) None of these

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TIME SPEED AND DISTANCE

Q 5. A person goes to his in-laws house at 40 km/hr he reach 2 hrs early and next time he goes with 30 km/h, he reach 1 hour earlier. Find distance, actual time and speed?

(a) 120 km,5h,24km/h (b) 150km,6h,30km/h (c) 180km,8h,35km/h

(d) None of these

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TIME SPEED AND DISTANCE

Q 6. A man starts to travel in order to reach his in-laws house at the speed of 60 km/hr and reaches 1 hours late. If travels at 80 km/hr than he is reaches 2 hours early. What is the distance, actual speed and actual time?

(a) 720 km,11h,720/11km/h (b) 750km,6h,50km/h (c) 780km,8h,35km/h

(d) None of these

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TIME SPEED AND DISTANCE

Q 7. If a man runs at 10 km/h than he arrives a a certain place at 1 pm. But if he increases his speed 5 km/h than he reached there at 11 a.m. At what speed must he run to reaches there at 12 p.m?

(a) 8 km/hr (b) 16 km/hr (c) 12 km/hr (d) None of these

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(a) 8 km/hr (b) 16 km/hr (c) 12 km/hr (d) None of these

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TIME SPEED AND DISTANCE

Q 8. A person can travel a certain distance in 30 hours if he decrease his speed by $1/15^{th}$ than he would cover 10 km less in same time. Find the speed of that person.

(a) 8 km/hr (b) 6 km/hr (c) 5 km/hr (d) None of these

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TIME SPEED AND DISTANCE

Q 8. A person can travel a certain distance in 30 hours if he decrease his speed by $1/15^{th}$ than he would cover 10 km less in same time. Find the speed of that person.

(a) 8 km/hr (b) 6 km/hr (c) 5 km/hr (d) None of these

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TIME SPEED AND DISTANCE

Q 9. A policeman saw a thief at a 700 metre distance start to chase. Policeman can run 47 km in 34 min and the thief can run 47 km in 41 min. Find the distance travelled by thief before arrested?

(a) 3.4 km (b) 1.6 km/hr (c) 1.2 km/hr (d) None of these

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TIME SPEED AND DISTANCE

Q 9. A policeman saw a thief at a 700 metre distance start to chase. Policeman can run 47 km in 34 min and the thief can run 47 km in 41 min. Find the distance travelled by thief before arrested?

(a) 3.4 km (b) 1.6 km/hr (c) 1.2 km/hr (d) None of these

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TIME SPEED AND DISTANCE

Q 10. Distance b/w A and B is 60 km. Two person P and Q start from A simultaneously to pt B speed of P is 4 km/h less than speed of Q. Q reached at B and return immediately and meet P a distance of 12 k.m from B. Find speed of P and Q?

(a) 8 & 10 km/hr (b) 8 & 12 km/hr (c) 12 & 12 km/hr

(d) None of these

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TIME SPEED AND DISTANCE

Q 11. Radha and Shyam with the speed of 15 km/hr and 21 km/h start to meet each other. At the time of meeting shyam has travelled 500 metre more than Radha. Find the distance b/w Radha and Shyam?

(a) 3 km (b) 6 km (c) 12 km (d) None of these

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TIME SPEED AND DISTANCE

Q 12. Radha and Shyam start to meet each other one from Agra and Shyam from mothura speed of each is 5 km/hr. Radha decrease her speed one k.m per hour after every hour. And shyam increases his speed 1 km/hr after every 1 hour. If the distance b/w Agra and Mathura is 110 km. Find the time of meeting?

(a) 8 hr (b) 10 hr (c) 12 hr (d) None of these

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(a) 8 hr (b) 10 hr (c) 12 hr (d) None of these

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TIME SPEED AND DISTANCE

Q 13. If a boy increases his speed by 3 km/hr than he reach 50 min. Early. If he decreases his speed by 2 km/hr then he reached 40 min. Late. Find the distance, actual speed and time?

(a) 27km/hr,500min,225km

(b) 36km/hr,600min,300km

(c) 42km/hr,400min,250km

(d) None of these

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TIME SPEED AND DISTANCE

Q 14. A person has to cover 360 km distance. If he increases his speed by 10 km/h. He reaches 3 hours early. Find his initial speed?

(a) 18 km/hr (b) 26 km/hr (c) 30 km/hr (d) None of these

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TIME SPEED AND DISTANCE

Q 15. Two bullets are fired from a place at an interval of 11 min. A person approaching to that place hears it at the gap of 10 min. If speed of sound is 330 m/sec. then what is the speed of person.

(a) 33 m/s (b) 25 m/s (c) 20 m/s (d) None of these

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TIME SPEED AND DISTANCE

Q 16. Two bom bs blast at an interval of 10 Min. A person running away from that place hears the two blast at an interval of 12 min. If speed of sound is 330 m/sec. Find the speed of person and the distance covered by the person running away from the blast.

(a) 45 m/s (b) 50 m/s (c) 55 m/s (d) None of these

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TIME SPEED AND DISTANCE

Q 17. A person cover a distance of 300 km partly by train and partly by car. If he travels 60km by train and rest by car it takes him 4 hours to cover that distance. If he covers 100 km by train and rest by car it takes him 4 hours 10 min. To cover that distance. Find out the speed of train and car?

(a) 60,80 km/hr (b) 60,70 km/hr (c) 60,90 km/hr (d) None of these

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TIME SPEED AND DISTANCE

Q 18. **A carriage driving in a fog passed a man who was walking at the rate of 6 km an hour in the same direction. He could see the carriage for 8 minutes and it was visible to him upto a distance of 200 m. What was the speed of the carriage?**

(a) 9 km/hr (b) 4.5 km/hr (c) 7.5 km/hr
(d) 8 km/hr

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Q 19. After travelling a distance of 50km has meets with an accident and its speed becomes 3/4th of its actual speed and take 35 min. more if the accident had occurred after travelling 24 km more than the train would have reached the station 25 min. late. Find out speed of train?

(a) 48 km/hr (b) 56 km/hr (c) 72 km/hr (d) None of these

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Q 20. After travelling a distance of 50km has meets with an accident and its speed becomes 3/4th of its actual speed and take 35 min. more if the accident had occurred after travelling 24 km more than the train would have reached the station 25 min. late. Find out distance travel by train?

(a) 120 km (b) 134 km (c) 150 km (d) None of these

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TIME SPEED AND DISTANCE

Q 21. A train met with an accident 60 km away from Anantpur station. It completed the remaining journey at 5/6th of the previous speed and reached the Baramula station 1 hour 12 min. late. Had the accident take place 60 km further, it would have been only 1 hour late. What is the distance between Anantpur and Baramula?

(a) 420 km (b) 650 km (c) 720 km (d) None of these

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THANK YOU

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